ECE 310 Fall 2026
Lecture 19: Aliasing Effect

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Learning Objectives

After this lecture, you should be able to:

- Define and identify the aliasing effect when sampling bandlimited continuous-time signals using Nyquist criterion.
- Demonstrate how sampling below the Nyquist rate aliases sinusoidal signals in the time-domain and frequency-domain.



Nyquist Criterion

Nyquist criterion

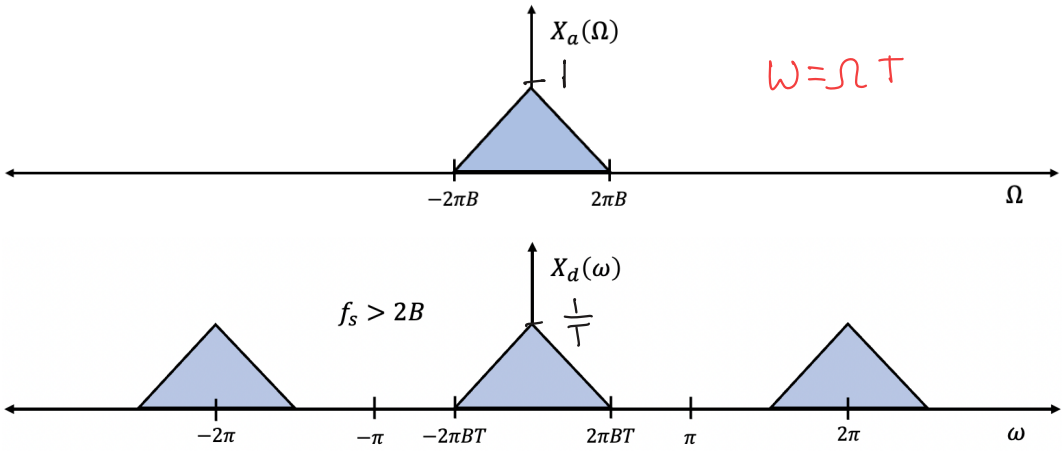
A bandlimited continuous-time signal with maximum linear frequency B can be sampled at sampling period T to produce $x[n]$ such that $x_a(t)$ may be recovered from $x[n]$ if

$$T < \frac{1}{2B}$$
$$f_s > 2B$$

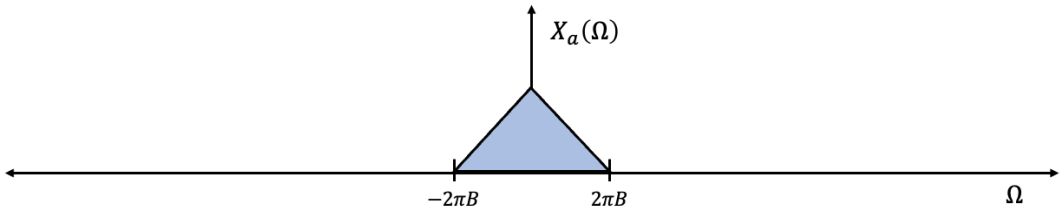
Otherwise, we will incur *aliasing* when obtaining $x[n]$. The frequency $f_{\text{nyquist}} = 2B$ (units in Hz) is known as the *Nyquist rate*.

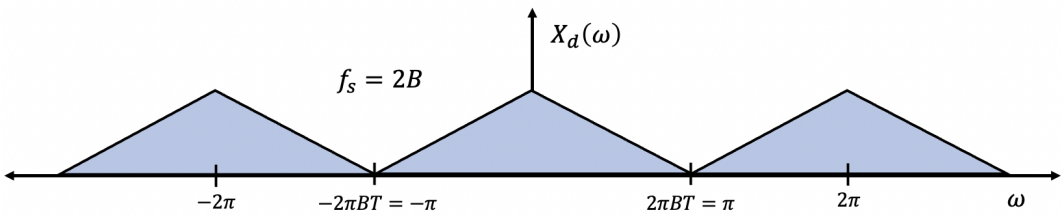


Nyquist Criterion

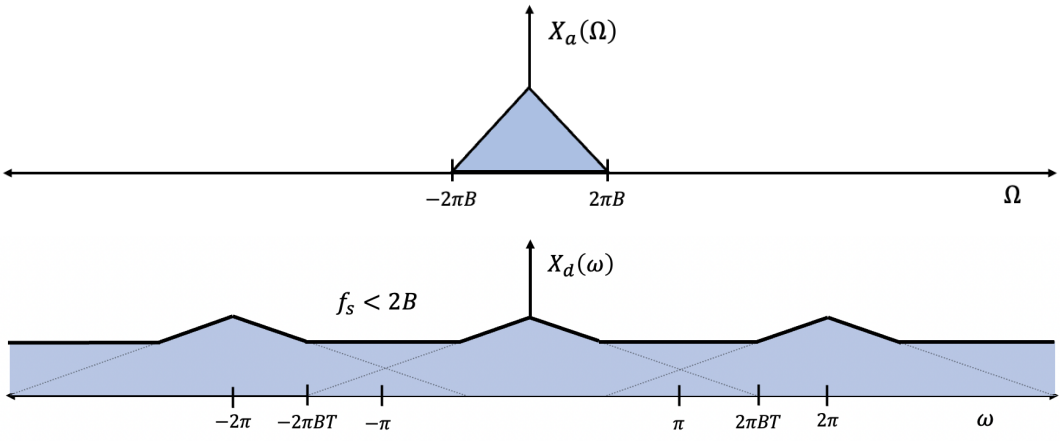


Nyquist Criterion





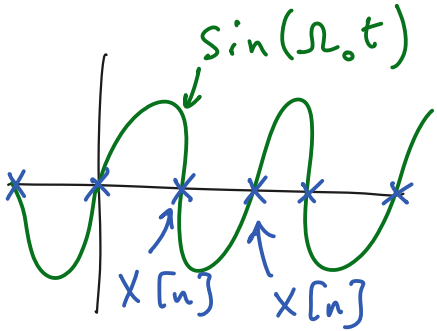
Nyquist Criterion



Can we sample *at* the Nyquist rate?

Consider $x_a(t) = \sin(\Omega_0 t)$ sampled at $f_s = f_{\text{nyquist}} = 2B = \frac{\Omega_0}{\pi}$ seconds.

$$
\begin{aligned}
X[n] &= x_a(nT) \\
&= \sin(\Omega_0 nT) \\
&= \sin(\pi n) \\
&= 0 \quad \text{for all } n
\end{aligned}
$$



Aliasing

Aliasing

- Bandlimited analog signals $x_a(t)$ sampled *above* the Nyquist rate may be perfectly recovered from $x[n]$ via ideal digital-to-analog conversion.
- Otherwise, i.e. sampling below the Nyquist rate, the sampled signal $x[n]$ will have an incorrect, alternative appearance or *alias*. The resulting $x[n]$ is said to exhibit *aliasing*.



Aliasing Example

Consider an analog signal $x_a(t) = \sin(100\pi t)$. This signal has $B = f_{\max} = 50$ Hz. Thus, the Nyquist sampling rate is $f_{\text{nyquist}} = 100$ Hz.

We will consider the result of sampling $x_a(t)$ at the following sampling rates:

1. $f_s = 160$ Hz, $T = \frac{1}{160}$ s. (Well above Nyquist)
2. $f_s = 120$ Hz, $T = \frac{1}{120}$ s. (Slightly above Nyquist)
3. $f_s = 80$ Hz, $T = \frac{1}{80}$ s. (Slightly below Nyquist)
4. $f_s = 40$ Hz, $T = \frac{1}{40}$ s. (Well below Nyquist)

Finally, recall

$$\sin(\omega_0 n) \overset{\mathcal{F}}{\longleftrightarrow} -j\pi \left(\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right)$$



Aliasing Example: Well Above Nyquist

$x_a(t) = \sin(100\pi t),\ f_s = 160$ Hz, $T = \frac{1}{160}$ s.



Aliasing Example: Slightly Above Nyquist

$x_a(t) = \sin(100\pi t),\ f_s = 120$ Hz, $T = \frac{1}{120}$ s.



Aliasing Example: Slightly Below Nyquist

$x_a(t) = \sin(100\pi t),\ f_s = 80$ Hz, $T = \frac{1}{80}$ s.

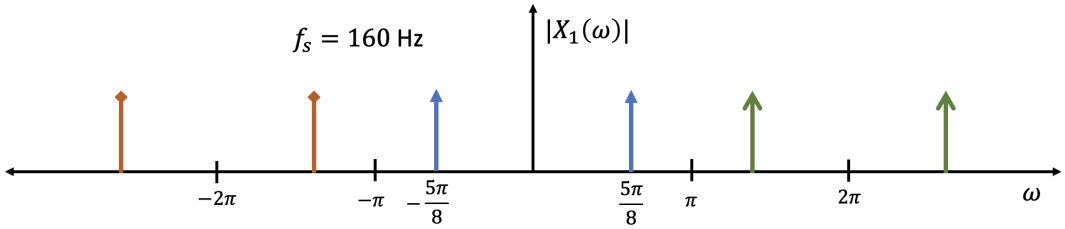


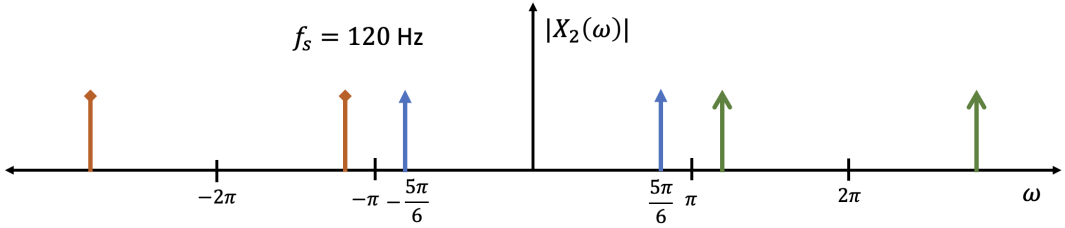
Aliasing Example: Well Below Nyquist

$x_a(t) = \sin(100\pi t),\ f_s = 40$ Hz, $T = \frac{1}{40}$ s.

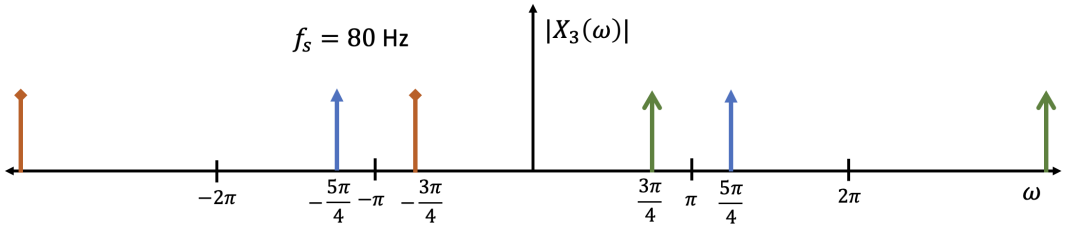


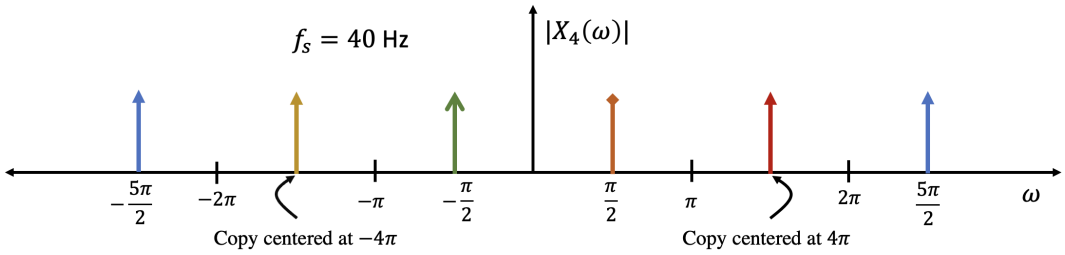
Aliasing Example: Frequency Domain



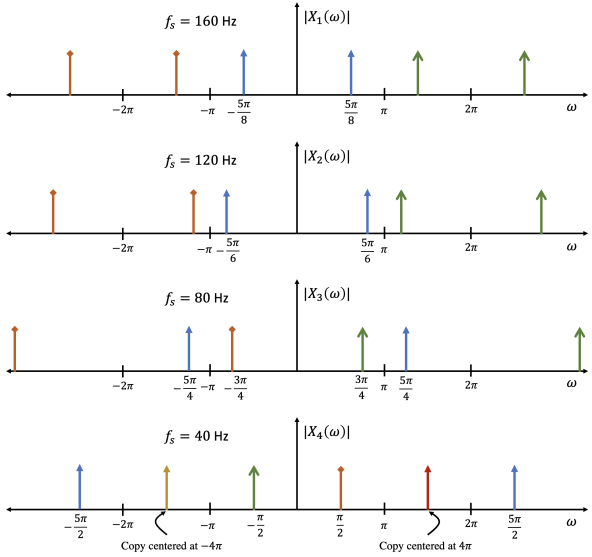


Aliasing Example: Frequency Domain

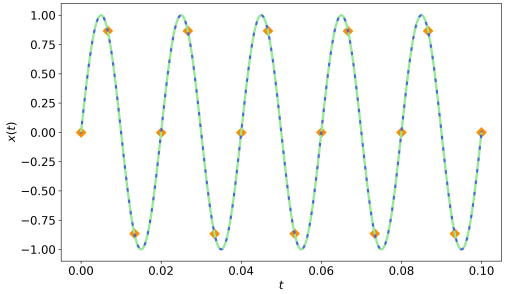




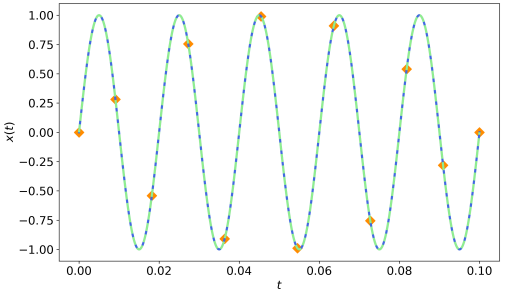
Aliasing Example: Frequency Domain



Aliasing Example: Time Domain

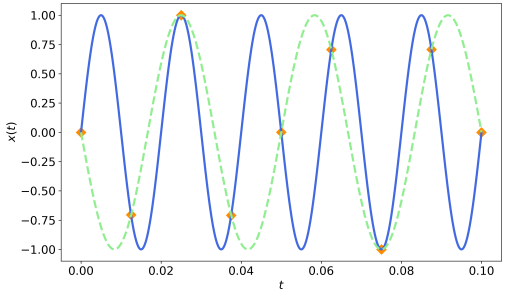


(a) $f_s = 160$ Hz

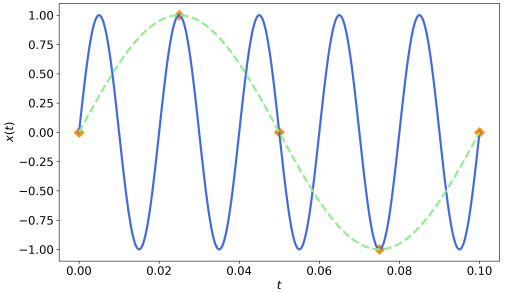


(b) $f_s = 120$ Hz

Aliasing Example: Time Domain



(a) $f_s = 80$ Hz



(b) $f_s = 40$ Hz

Aliasing Example: Possible Frequencies

Suppose continuous-time signal $x_a(t) = \cos(\Omega_0 t)$ is sampled at f_s=100 Hz for some frequency $\Omega_0 \in \mathbb{R}$ and the resulting discrete-time signal is $x[n] = \cos\left(\frac{\pi}{3}n\right)$. Determine two possible choices of Ω_0 consistent with this information. **Hint:** Remember to consider aliasing!

